Bt302: Immunology

Spring 2021

Total Marks: 10

Due date: 26th July 2021

Questions

1. Describe hypersensitivity. Write comparison of types of hypersensitivity in detail.

Hypersensitivity can be defined as a condition in which the immune system of the body has a negative effect on the body. It can be in different ways like allergy, it is an abnormal immunological response again the stimulus of the environment.

Types of Hypersensitivity:

- Type I (Anaphylactic)
- Type II (Cytotoxic)
- Type III (Immune complex)
- Type IV (delayed cell mediated)

Type I (Anaphylactic)

Happens in <30 min.

- IgE is involved which binds to mast cells
- Mast cells degranulate and secretes mediators of inflammation.
- Histamine, leukotrienes, PG
- Shock from drug reactions
- Venoms
- Common allergens causing asthma Diagnosis by demonstration of IgE by i/d injections of allergens one by one.
- Quantitation of IgE by ELISA
- Rx: Epinephrine & antihistaminic drugs

Type II (Cytotoxic)

Involves

- Antigenic cell

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 IgG or IgM

 Complement

 Blood transfusion reaction is an example.

Diagnosis

 Cell deficiencies

 Low number of RBCs due to hemolysis

 Thrombocytopenia if platelets are involved.

Rx

Avoid those drugs that cause it.

 Blood transfusion with correct blood type

 Symptomatic treatment

Type III (Immune complex)

 Involves

 Soluble Antigen circulating in the serum

 IgG or sometimes IgM

 Complement

Diagnosis by history and symptoms

 Glomerulonephritis

 Confirmed by biopsy and testing of presence of IgG in kidney cells

 Rx. Corticosteroids

Type IV (delayed cell mediated)

• Originally called

tuberculin reaction

• Robert Koch performed the test.

• Contact dermatitis is the most common example.

Involves:

– Antigen

– T cells

– Macrophages

• Transplant rejection is good example.

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• Reaction to hair dye

Diagnosis:

– Skin testing

• Rx: symptomatic

• Avoid contact

2. Write mechanism of immune response against intracellar and extracellular pathogen? 5

Mechanism against Intracellular Pathogen

Intracellular bacteria are endowed with the capacity to survive and replicate inside mononuclear phagocytes (MP) and sometimes, within certain other host cells. MP are potent effectors cells which are able to engulf and kill many bacterial invaders. Therefore, intracellular bacteria had to exploit potent evasion mechanisms that allow their survival in this hostile environment. At the early phase, natural killer cells activate antibacterial defence mechanisms. During intracellular persistence, microbial proteins are processed and presented, thus initiating T cell activation. By secreting interleukins, CD4 alpha/beta TH1 cells activate MP, converting them from a habitat to a potent effector cell; TH2-dependent activities seem to be of minor importance. Cytolytic CD8 T cells represent a further element of protection. In the case of Listeria monocytogenes, the gene products responsible for virulence and for the introduction of antigens into the MHC class I pathway are being characterized. Increasing evidence points to a role of gamma/delta T lymphocytes in antibacterial immunity, although their precise function remains to be elucidated. Protection in the host is a local event focussed on granulomatous lesions. MP accumulate at the site of microbial growth and become activated through the CD4 T cell-interleukin-MP axis. Lysis of incapacitated MP and other host cells by CD8 T cells allows release and subsequent uptake by more efficient phagocytes, thus contributing to host protection.

Mechanism against extracellular pathogens

In an infection by extracellular bacteria, the host triggers a series of responses to combat the pathogen and prevent its spread. The main mechanism of the innate immune response to

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eradicate bacteria is activation of the complement system, phagocytosis, and inflammatory response.

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